

Electronic Supplementary Material

Early physiological phenotypes and outcomes in critically ill adults with non-traumatic subarachnoid hemorrhage

This Electronic Supplementary Material (ESM) accompanies the Intensive Care Medicine-style main manuscript. It provides cohort algorithms, code lists, variable mappings, extended tables, sensitivity analyses, supplementary figures, and reproducibility details.

ESM 1. Cohort Algorithm and Provenance

ESM Table 1. MIMIC-IV cohort-flow counts

Step	Definition	Admissions	Patients
01	Source SAH admissions	1,576	1,531
02	Adult source SAH	1,576	1,531
03	Adult non-traumatic SAH; no aneurysm evidence required	suppressed (transition <10)	suppressed (transition <10)
04	First ICU stay among NSAH hospitalizations	1,317	1,298
05	ICU length of stay ≥ 24 h	1,212	1,197
06	Eight core features with ≤ 2 missing values	1,186	1,173
07	Primary analysis after excluding massive early transfusion	1,186	1,173
08	Sensitivity cohort: ICU length of stay ≥ 48 h	1,005	996
09	Sensitivity cohort: no recorded RBC transfusion in 0–48 h	1,162	1,149

ESM Table 2. Historical eICU MIMIC-anchored cohort-flow counts

Step	Definition	Unit stays	Patients
01	SAH candidate unit stays from diagnosis/admissionDx	2,880	2,571
02	Adult patients	2,863	2,554
03	Exclude traumatic SAH flags	1,314	1,158
04	First ICU stay per unique patient	suppressed (patient transition <10)	suppressed (patient transition <10)
05	ICU length of stay ≥ 24 h	903	903
06	Eight core features with ≤ 2 missing values	843	843
07	Sensitivity cohort: ICU length of stay ≥ 48 h	626	626
08	Sensitivity cohort: no recorded RBC transfusion in 0–48 h	819	819
09	Strict SAH sensitivity definition	605	605
10	Low-missing sensitivity: ≤ 1 missing core feature	819	819
11	Complete-case sensitivity	428	428

Counts in ESM Tables 1 and 2 are derived from `dist/figures/fig1_cohort_flowchart_data.json`, which was generated from BigQuery intermediate cohort-flow tables.

ESM Table 3. Diagnostic code and text algorithm

Data source	Inclusion evidence	Exclusion evidence
MIMIC-IV	ICD-9 <code>430</code> (subarachnoid hemorrhage); ICD-10 <code>I60.x</code> (nontraumatic subarachnoid hemorrhage subtypes)	ICD-10 <code>S06.6x</code> or diagnostic text consistent with traumatic subarachnoid hemorrhage
eICU-CRD	ICD-9 <code>430</code> ; <code>apacheadmissiondx</code> containing “subarachnoid	Diagnosis/admission text consistent with traumatic subarachnoid hemorrhage

Data source	Inclusion evidence	Exclusion evidence
	hemorrhage”; diagnosis table text consistent with subarachnoid hemorrhage	

The algorithm intentionally did not require aneurysm diagnosis or aneurysm-securing procedure evidence for the primary NSAH cohort. A strict aneurysm subgroup was evaluated as sensitivity analysis.

ESM 2. Variable Mapping and Preprocessing

ESM Table 4. Core physiological feature definitions

Domain	Feature	Definition	Window	Primary preprocessing
Neurological	gcs_motor_min_48h	Minimum GCS motor score	0–48 h after ICU admission	Z-score standardization
Circulatory	map_min_48h	Minimum mean arterial pressure	0–48 h after ICU admission	Z-score standardization
Circulatory	shock_index_max_48h	Maximum heart rate / systolic blood pressure	0–48 h after ICU admission	Z-score standardization
Oxygenation	spo2_min_48h	Minimum pulse oximetry oxygen saturation	0–48 h after ICU admission	Z-score standardization
Renal	creatinine_max_48h	Maximum serum creatinine	0–48 h after ICU admission	log1p transformation, then Z-score
Coagulation	inr_max_48h	Maximum international normalized ratio	0–48 h after ICU admission	log1p transformation, then Z-score
Hematologic	hb_min_48h_all	Minimum hemoglobin	0–48 h after ICU admission	Z-score standardization
Hematologic	platelet_min_48h	Minimum platelet count	0–48 h after ICU admission	Z-score standardization

Missing values were imputed using MIMIC-IV derivation-cohort medians. The historical exploratory eICU analysis reconstructed MIMIC-IV medians, scaler parameters, PCA loadings, and K-means centroids during the validation run. The

exact serialized transform was not retained in the public reproducibility bundle, so authorized re-execution under the new fail-closed artifact contract remains pending.

ESM Table 5. Feature availability and data quality

Feature	MIMIC missing N (N=1,186)	MIMIC missing %	eICU missing N (N=903 before feature-completeness eligibility)	eICU missing %
inr_max_48h	65	5.48%	446	49.4%
creatinine_max_48h	1	0.08%	31	3.4%
platelet_min_48h	0	0.00%	75	8.3%
hb_min_48h_all	0	0.00%	64	7.1%
shock_index_max_48h	0	0.00%	10	1.1%
map_min_48h	0	0.00%	10	1.1%
spo2_min_48h	0	0.00%	10	1.1%
gcs_motor_min_48h	0	0.00%	5	0.6%

Among the 843 patients who passed eICU feature-completeness eligibility, 390 (46.3%) had missing INR and received a MIMIC-derived median in the historical reconstruction. The INR-free sensitivity expanded eligibility to 868 patients.

ESM Table 6. Principal component loadings for the standardized feature space

Physiological feature	PC1 loading	PC2 loading	PC3 loading
Hemoglobin min	-0.340	-0.430	-0.100
GCS motor min	-0.370	0.020	-0.490
MAP min	-0.300	0.410	-0.250
Shock index max	0.410	-0.390	0.260

Physiological feature	PC1 loading	PC2 loading	PC3 loading
SpO2 min	-0.190	0.510	0.670
Creatinine max	0.410	0.080	-0.190
INR max	0.420	0.170	-0.290
Platelets min	-0.310	-0.450	0.230

ESM 3. Extended Tables

ESM Table 7. Baseline characteristics of the MIMIC-IV derivation cohort

Characteristic	Overall (N=1,186)	P1 (n=694)	P2 (n=384)	P3 (n=108)	Unadjusted p-value
Demographics					
Age, median [IQR], years	61.0 [50.0–71.0]	59.5 [49.0–70.0]	62.0 [52.0–75.0]	62.5 [48.0–70.0]	0.026
Female sex, n (%)	665 (56.1%)	393 (56.6%)	234 (60.9%)	38 (35.2%)	<0.001
White race, n (%)	696 (58.7%)	449 (64.7%)	191 (49.7%)	56 (51.9%)	<0.001
Black race, n (%)	89 (7.5%)	50 (7.2%)	29 (7.6%)	10 (9.3%)	
Other race, n (%)	173 (14.6%)	95 (13.7%)	56 (14.6%)	22 (20.4%)	
Unknown race, n (%)	228 (19.2%)	100 (14.4%)	108 (28.1%)	20 (18.5%)	
Admission details					
Emergency admission, n (%)	830 (70.0%)	469 (67.6%)	307 (79.9%)	54 (50.0%)	<0.001
Observation admission, n (%)	190 (16.0%)	140 (20.2%)	35 (9.1%)	15 (13.9%)	
Urgent admission, n (%)	121 (10.2%)	66 (9.5%)	26 (6.8%)	29 (26.9%)	
Surgical same-day admission, n (%)	32 (2.7%)	15 (2.2%)	12 (3.1%)	<10	
Direct emergency / other, n (%)	13 (1.1%)	<10	<10	<10	

Characteristic	Overall (N=1,186)	P1 (n=694)	P2 (n=384)	P3 (n=108)	Unadjusted p-value
Etiology and evidence					
NSAH evidence level 1, n (%)	423 (35.7%)	252 (36.3%)	99 (25.8%)	72 (66.7%)	<0.001
NSAH evidence level 2, n (%)	529 (44.6%)	305 (43.9%)	190 (49.5%)	34 (31.5%)	
NSAH evidence level 3, n (%)	234 (19.7%)	137 (19.7%)	95 (24.7%)	<10	
Aneurysm diagnosis, n (%)	87 (7.3%)	53 (7.6%)	31 (8.1%)	<10	0.157
Aneurysm-securing procedure, n (%)	234 (19.7%)	137 (19.7%)	95 (24.7%)	<10	<0.001
ICU interventions (0–48 h)					
Nimodipine, n (%)	675 (56.9%)	432 (62.2%)	226 (58.9%)	17 (15.7%)	<0.001
EVD or ICP monitoring, n (%)	322 (27.2%)	119 (17.1%)	188 (49.0%)	15 (13.9%)	<0.001
Vasopressor use, n (%)	252 (21.2%)	43 (6.2%)	154 (40.1%)	55 (50.9%)	<0.001
Mechanical ventilation, n (%)	431 (36.3%)	120 (17.3%)	249 (64.8%)	62 (57.4%)	<0.001
Fluid balance, median [IQR], L	1.38 [-0.45–2.96]	0.94 [-0.65–2.26]	2.26 [0.22–3.79]	2.41 [-0.04–6.19]	<0.001
CRRT, n (%)	10 (0.8%)	0 (0.0%)	<10	<10	<0.001
Outcomes					
In-hospital mortality, n (%)	235 (19.81%)	44 (6.34%)	125 (32.55%)	66 (61.11%)	<0.001
ICU mortality, n (%)	182 (15.35%)	25 (3.60%)	102 (26.56%)	55 (50.93%)	<0.001
Early anemia (0–48 h), n (%)	315 (26.56%)	84 (12.10%)	159 (41.41%)	72 (66.67%)	<0.001
RBC transfusion (0–48 h), n (%)	24 (2.02%)	<10	13 (3.39%)	<10	<0.001
ICU LOS, median [IQR], days	6.96 [2.88–13.41]	6.38 [2.71–10.50]	10.56 [3.83–17.50]	5.38 [2.10–12.19]	<0.001
Hospital LOS, median [IQR], days	11.60 [6.42–19.91]	9.94 [6.51–15.24]	15.85 [6.70–24.79]	13.31 [3.46–28.29]	<0.001

ESM Table 8. Early physiological profiles of identified phenotypes

Physiological variable (0–48 h)	Overall (N=1,186)	P1 (n=694)	P2 (n=384)	P3 (n=108)	Unadjusted p-value
Hemoglobin min, g/dL	11.10 [9.80–12.40]	11.70 [10.80–12.80]	10.25 [9.10–11.50]	8.85 [7.38–10.43]	<0.001
GCS motor min	5.00 [1.00–6.00]	6.00 [5.00–6.00]	1.00 [1.00–4.00]	1.00 [1.00–5.00]	<0.001
MAP min, mmHg	61.00 [54.00–67.00]	64.00 [58.00–70.00]	58.00 [51.00–63.00]	55.00 [47.50–61.00]	<0.001
Shock index max	0.88 [0.77–1.05]	0.82 [0.73–0.92]	1.00 [0.87–1.17]	1.16 [0.93–1.42]	<0.001
SpO2 min, %	92.00 [90.00–94.00]	92.00 [90.00–94.00]	93.50 [92.00–96.00]	88.50 [78.00–92.00]	<0.001
Creatinine max, mg/dL	0.80 [0.70–1.10]	0.80 [0.60–0.90]	0.90 [0.70–1.10]	1.90 [1.18–3.20]	<0.001
INR max	1.20 [1.10–1.30]	1.10 [1.10–1.20]	1.20 [1.10–1.30]	1.80 [1.40–2.30]	<0.001
Platelets min, $\times 10^3/\mu\text{L}$	188.0 [150.0–232.0]	204.0 [169.25–245.0]	174.0 [140.0–213.5]	92.0 [40.75–160.25]	<0.001

ESM Table 9. Implemented exploratory logistic regression for in-hospital mortality

Variable	aOR (95% CI)	Unadjusted p-value
Phenotype 2 vs P1	7.59 (5.07–11.36)	<0.001
Phenotype 3 vs P1	21.21 (12.08–37.26)	<0.001
Early anemia	0.99 (0.68–1.44)	0.955
Age, per year	1.03 (1.02–1.04)	<0.001
Male sex	1.21 (0.86–1.70)	0.274
Aneurysm diagnosis	0.52 (0.24–1.14)	0.101
NSAH evidence level 2	0.88 (0.60–1.29)	0.499
NSAH evidence level 3	0.65 (0.38–1.09)	0.102
Emergency admission	2.13 (0.75–6.06)	0.156
Observation admission	1.81 (0.59–5.54)	0.298

Variable	aOR (95% CI)	Unadjusted p-value
Urgent admission	3.38 (1.11–10.33)	0.033

The implemented model included phenotype, age, sex, admission type, NSAH evidence level, aneurysm diagnosis, and early anemia. The formula was documented after outcome access. Reported p-values were not adjusted for multiplicity, and confidence intervals used stay-level model covariance rather than `subject_id`-clustered covariance. A frozen process-of-care model is retained in the reproducibility output but is not presented as an inferential result: it included same-window treatment variables, NSAH evidence level, and an aneurysm-securing indicator that exactly duplicated evidence level 3, leaving those etiology coefficients non-estimable.

ESM Note 1. Time-to-event analysis boundary

Cox estimates are not reported because phenotype assignment uses measurements collected during 0-48 h while deaths can occur during that same interval. Starting follow-up at ICU admission would therefore use future-informed phenotype assignments, and treating live discharge as non-informative censoring would not address the competing-event structure. The submitted analysis is limited to descriptive associations with in-hospital mortality. A future survival analysis would require a fixed landmark after phenotype ascertainment and an explicitly defined competing-risk estimand.

ESM Table 10. Historical exploratory eICU MIMIC-anchored cohort characteristics and outcomes

Characteristic	Overall (N=843)	P1 (n=540)	P2 (n=221)	P3 (n=82)
Age, median, years	60.0	59.0	61.0	63.0
Hemoglobin min, g/dL	11.4	11.9	10.7	9.5
GCS motor min	6.0	6.0	4.0	4.0
MAP min, mmHg	58.0	62.0	51.0	51.5
	0.97	0.88	1.14	1.26

Characteristic	Overall (N=843)	P1 (n=540)	P2 (n=221)	P3 (n=82)
Shock index max				
SpO2 min, %	90.0	90.0	92.0	74.5
Creatinine max, mg/dL	0.81	0.75	0.81	1.54
INR max	1.1	1.1	1.1	1.5
Platelets min, ×10 ³ /μL	200.0	207.0	203.0	146.0
APACHE score, median	43.0	36.0	57.5	79.0
APS score, median	33.0	27.0	49.0	67.0
Predicted hospital mortality, median	0.112	0.069	0.243	0.426
Early anemia (0–48 h), n (%)	183 (21.7%)	60 (11.4%)	76 (34.7%)	47 (58.0%)
RBC transfusion (0–48 h), n (%)	24 (2.8%)	<10	13 (5.9%)	<10
Hospital mortality, n (%)	121 (14.35%)	29 (5.37%)	57 (25.79%)	35 (42.68%)
ICU mortality, n (%)	75 (8.90%)	15 (2.78%)	36 (16.29%)	24 (29.27%)

ESM Table 11. Sensitivity analysis summary

Analysis	N	P1 mortality	P2 mortality	P3/P4 mortality	Minimum subgroup N
MIMIC primary log-PCA	1,186	6.34%	32.55%	61.11% (P3)	108
Historical eICU MIMIC- anchored assignment	843	5.37%	25.79%	42.68% (P3)	82
MIMIC complete case	1,120	6.53%	38.58%	65.07% (P3)	63
MIMIC strict aneurysm	763	5.73%	30.82%	50.00% (P3)	48

Analysis	N	P1 mortality	P2 mortality	P3/P4 mortality	Minimum subgroup N
MIMIC ICU LOS ≥ 48 h	1,005	5.96%	29.23%	58.06% (P3)	93
MIMIC no RBC 0–48 h	1,162	6.22%	31.28%	61.95% (P3)	113
MIMIC include massive transfusion	1,186	6.34%	32.55%	61.11% (P3)	108
MIMIC Hb-free clustering	1,186	6.85%	39.52%	64.10% (P3)	78
MIMIC INR-free clustering	1,186	6.00%	35.32%	66.67% (P3)	84
MIMIC K=4 resolution	1,186	6.17% (P1)	37.99% (P2)	61.54% (P3) / 56.41% (P4)	39
eICU ICU LOS ≥ 48 h	626	6.9%	24.2%	33.9% (P3)	62
eICU no RBC 0–48 h	819	5.4%	25.8%	39.7% (P3)	73
eICU strict SAH	605	6.6%	30.5%	45.6% (P3)	68
eICU INR-free transport	868	4.6%	24.4%	38.7% (P3)	124

ESM Table 11A. Sensitivity of the adjusted anemia-mortality association to phenotype specification

Phenotype specification used for adjustment	Early-anemia aOR (95% CI)	Unadjusted p-value	Interpretation
Primary phenotypes including minimum hemoglobin	0.99 (0.68–1.44)	0.955	Potential overadjustment because hemoglobin informed both phenotype assignment and anemia status
	1.54 (1.06–2.22)	0.022	

Phenotype specification used for adjustment	Early-anemia aOR (95% CI)	Unadjusted p-value	Interpretation
Phenotypes re-derived without hemoglobin			Association differed after removal of hemoglobin from phenotype derivation

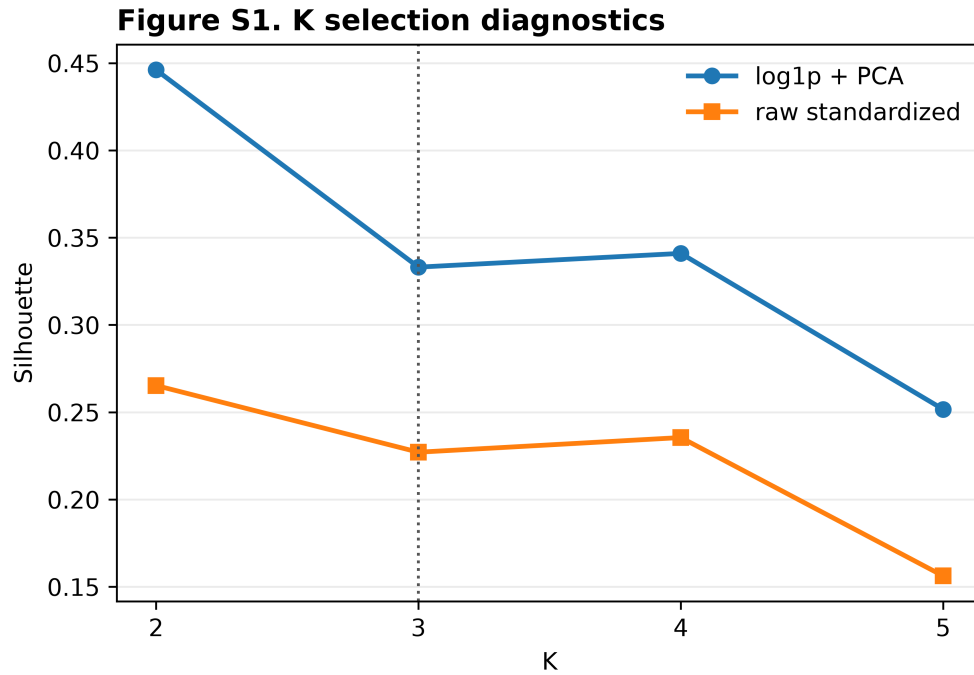
Both models included early anemia and adjusted for phenotype, age, sex, admission type, NSAH evidence level, and aneurysm diagnosis; phenotype assignment differed as shown in the first column. These post-outcome exploratory estimates used stay-level model covariance, and p-values were not adjusted for multiplicity. The differing estimates indicate sensitivity to phenotype specification and do not establish an independent prognostic or causal effect of anemia.

ESM Table 12. Bootstrap overall partition stability

Metric	Mean	Standard deviation	Minimum	Maximum
Adjusted Rand Index (ARI)	0.8554	0.0695	0.6015	0.9692
Same ordered label rate	94.93%	2.54%	85.67%	98.74%
Minimum cluster size, N	110.7	16.3	51.0	147.0

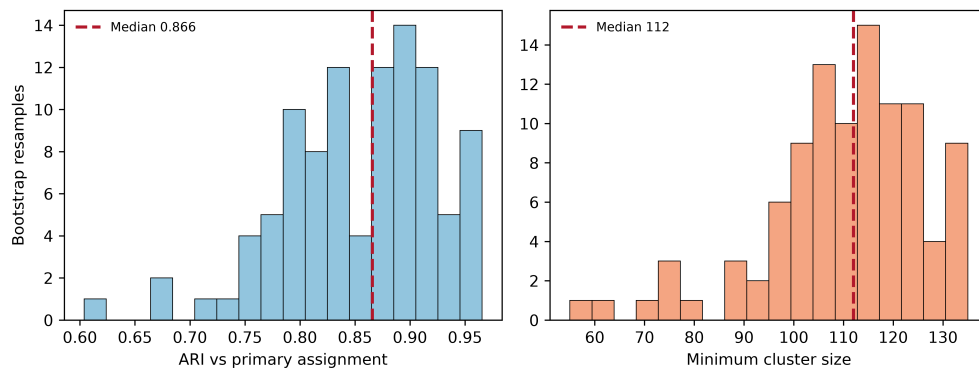
Cluster-specific Jaccard values, assignment-uncertainty distributions, and a multiple-seed summary were not preserved in the public frozen result and are therefore not assessed for this release. The reported ARI supports overall partition stability only.

ESM 4. Supplementary Figures

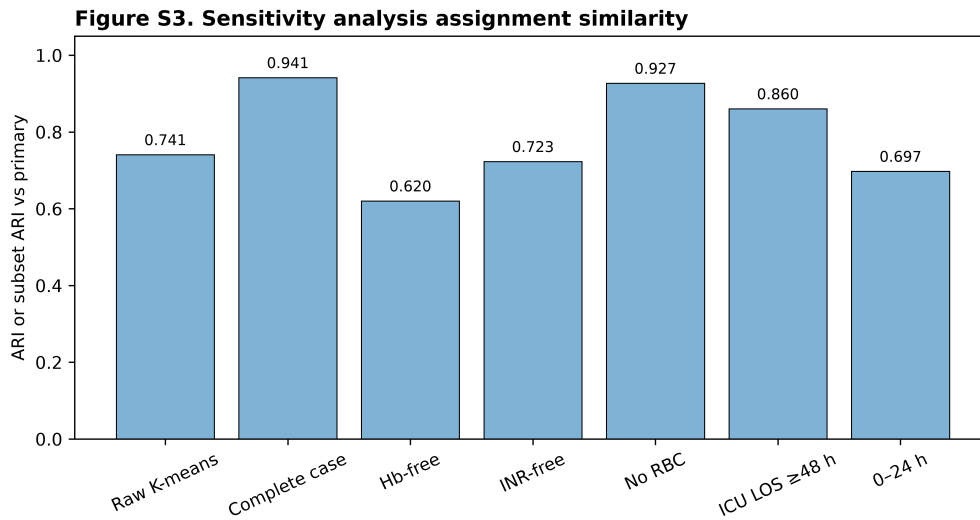


ESM Fig. 1. Cluster-quality metrics across K=2 to K=5 for raw standardized and primary log-PCA pipelines.

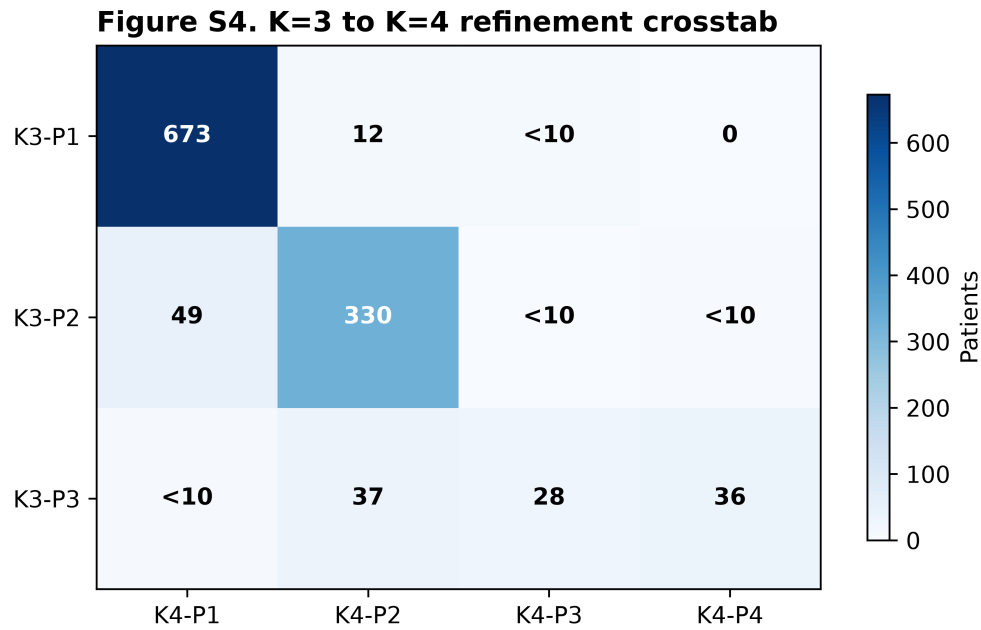
Figure S2. Bootstrap stability, 200 resamples



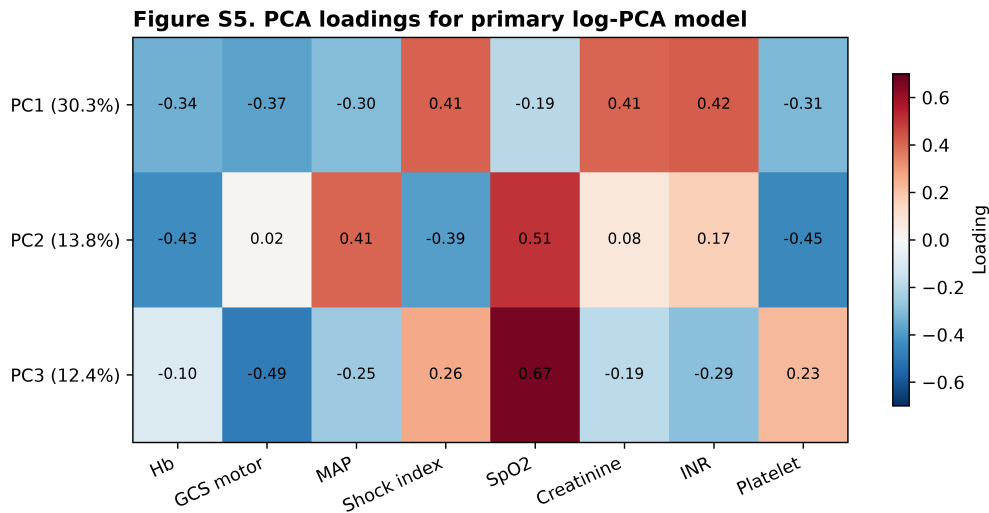
ESM Fig. 2. Subject-grouped, full-pipeline adjusted Rand Index distribution across 200 bootstrap iterations compared with the primary K-means K=3 partition.



ESM Fig. 3. Standardized cluster centers for sensitivity partitions, demonstrating alignment with the primary log-PCA centroids.

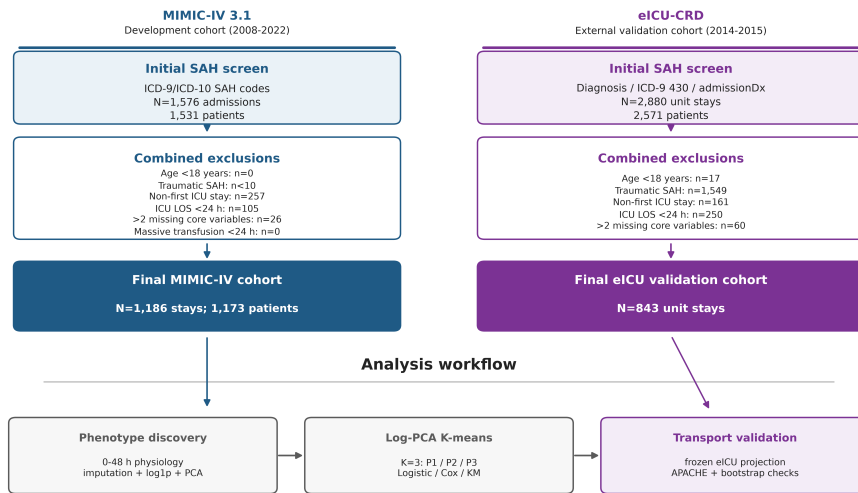


ESM Fig. 4. High-resolution K=4 refinement crosstab, illustrating how the primary P3 phenotype splits into two smaller high-risk subcohorts.



ESM Fig. 5. Loadings of the eight physiological variables on the first three principal components.

Cohort selection and analysis framework



Counts from BigQuery intermediate tables: non_traumatic_sah_studycohort_flowchart_counts and eicu_sah_validation.eicu_cohort_flowchart_counts. MIMIC was used for phenotype derivation; eICU was used for external transport validation.

ESM Fig. 6. Cohort selection and analysis design. Counts were derived from BigQuery intermediate cohort-flow tables. The lower panel summarizes MIMIC phenotype derivation and the historical exploratory eICU MIMIC-anchored reconstruction.

ESM 5. Reproducibility Notes

Analysis scripts

Step	Script or artifact	Purpose
Cohort SQL execution	<code>scripts/12_run_non_traumatic_sah_cohort_sql.py</code>	Runs <code>10_create_non_traumatic_sah_cohort.sql</code> in BigQuery
Phenotype analysis	<code>scripts/13_run_non_traumatic_sah_analysis.py</code>	Executes <code>11_bigquery_notebook_non_traumatic_sah_analysis.py</code> locally
Figure generation	<code>scripts/generate_manuscript_figures.py</code>	Regenerates 600 dpi PNG and vector PDF versions of all main and supplementary figures
PDF rendering	<code>scripts/convert_manuscript_to_pdf.py</code>	Converts manuscript and ESM markdown files to PDF
Cohort-flow counts	<code>dist/figures/fig1_cohort_flowchart_data.json</code>	Stores BigQuery-derived MIMIC and eICU flow counts

Key reproducibility parameters

Component	Value
Primary physiology window	0–48 h after ICU admission
Primary feature count	8 variables
Missing-data rule	≤2 missing core features
Imputation	MIMIC-IV derivation-cohort median; historically reconstructed for eICU, exact artifact pending authorized rerun
Skewed-feature transform	log1p for creatinine and INR
Standardization	MIMIC-IV derivation-cohort mean and standard deviation
PCA components retained	3
Variance explained	56.41%
K-means K	3
K-means random state	42
K-means n_init	100
Bootstrap iterations	200

Component	Value
External transport assessment	Historical MIMIC-anchored reconstruction; immutable transform artifact not retained, authorized rerun required

BigQuery provenance

The project-level BigQuery dataset for the current non-traumatic SAH study is `mimic-study-498508.non_traumatic_sah_study`. MIMIC-IV source tables were drawn from PhysioNet MIMIC-IV 3.1 datasets. eICU validation used the eICU-CRD source tables available through PhysioNet. The exact final table names should be verified after the final BigQuery rerun and recorded in the repository release notes before journal submission.